

Urban-attribute results across splits

Qwen3.5-9B (per-city & seen/unseen) — 4-way image ablation

Setup & what each split tests

Two LoRA-trained **Qwen3.5-9B** models, each evaluated on **its own validation split** and on the **shared benchmark**. Four image modes: **blind** (no images), **sv** (street-view only), **sat** (marked satellite only), **full** (both).

- **per-city val** — *unseen locations within cities the model trained on*. Tests in-distribution generalisation. ($n = 6,984$)
- **seen/unseen val** — *entirely held-out cities the model has **never seen at all***. The hard test: geographic generalisation to brand-new cities. ($n = 8,831$)
- **benchmark** — a disjoint set of cities *and* locations not in *either* training set; the common held-out yardstick for both models. ($n = 5,240$)

Fusion = full — max(sv, sat). All: thinking off, greedy, edge 512 (matches training). Bold sv/sat = better single view.

How to read fusion on these 9 tasks

The 9 urban-attribute tasks are, **by construction, answerable from a single view** — land use, building height, road type, etc. are each readable from *either* the satellite or the street-view alone. So we **expect** $\text{full} \approx \text{the better single view}$, and $\text{fusion} \approx 0$ here is the **designed behaviour, not a model deficiency**.

What these tables actually show:

- **full never loses** to the better single view ($\text{fusion} \gtrsim 0$ throughout) — adding the second view doesn't confuse the model.
- Where one view is uninformative, full correctly tracks the **informative** one (bolded sv/sat per row).
- Tasks that *do* require combining views (mismatch_^*) live in the appendix and fuse strongly (+40–70) — the mechanism works when the task demands it.

per-city — OWN validation split (seen cities, new locations)

task	blind	sv	sat	full	full—best1
Land use	68.6	76.6	74.5	75.2	-1.4
Building height	41.0	64.8	67.0	66.3	-0.8
Urban density	55.1	62.0	54.7	59.4	-2.7
Junction type	58.0	68.3	74.1	74.6	+0.4
Green space	100.0	100.0	100.0	100.0	+0.0
Amenity richness	30.3	57.6	54.7	62.4	+4.8
Road type	66.3	76.5	71.3	75.2	-1.2
Road surface	94.2	94.0	93.7	94.2	+0.3
Transit density	28.5	46.9	47.6	50.6	+3.0
OVERALL (all topics)	48.6	58.4	54.3	75.5	

seen/unseen — OWN validation split (never-seen cities)

task	blind	sv	sat	full	full – best1
Land use	65.9	69.3	68.2	69.1	-0.1
Building height	40.4	59.2	55.1	58.8	-0.4
Urban density	70.1	66.8	66.4	70.0	+3.1
Junction type	68.9	71.4	72.0	73.5	+1.5
Green space	100.0	99.1	98.7	96.0	-3.1
Amenity richness	26.1	54.1	48.7	54.0	-0.1
Road type	65.8	72.0	70.1	70.0	-2.0
Road surface	98.8	98.1	97.2	98.1	+0.0
Transit density	27.2	42.9	44.2	45.2	+0.9
OVERALL (all topics)	48.6	55.7	52.4	73.1	

These cities appear *nowhere* in training — accuracy here is pure geographic generalisation.

Both models on the SHARED benchmark — 9 tasks

task	per-city full	per-city fus	seen/uns full	seen/uns fus
Land use	75.3	-0.2	73.9	+1.4
Building height	66.8	-0.5	66.8	+0.5
Urban density	65.1	+2.4	74.6	+3.1
Junction type	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	98.0	-1.5
Amenity richness	58.7	+2.1	55.6	+0.0
Road type	76.5	-1.0	76.7	+0.0
Road surface	95.7	+0.7	95.0	-0.3
Transit density	50.1	+4.5	49.4	+3.6

Mean fusion over 9 tasks — per-city **+1.1**, seen/unseen **+1.0** (excl. leaked green-space: +1.2 / +1.3). Disjoint cities + locations from both training sets.

per-city — benchmark (all 4 modes)

task	blind	sv	sat	full	full—best1
Land use	68.6	75.1	75.5	75.3	-0.2
Building height	45.3	66.3	67.4	66.8	-0.5
Urban density	54.6	62.7	61.8	65.1	+2.4
Junction type	61.4	68.6	73.4	75.1	+1.8
Green space	100.0	100.0	100.0	100.0	+0.0
Amenity richness	26.4	56.5	53.4	58.7	+2.1
Road type	68.6	77.4	72.4	76.5	-1.0
Road surface	95.7	93.4	95.0	95.7	+0.7
Transit density	28.3	45.6	43.7	50.1	+4.5
OVERALL (all topics)	48.5	57.7	54.9	75.9	

seen/unseen — benchmark (all 4 modes)

task	blind	sv	sat	full	full—best1
Land use	63.4	72.4	72.0	73.9	+1.4
Building height	50.5	66.3	63.7	66.8	+0.5
Urban density	53.2	64.8	71.5	74.6	+3.1
Junction type	62.0	71.3	74.0	76.3	+2.4
Green space	100.0	99.5	99.0	98.0	-1.5
Amenity richness	25.7	55.6	53.0	55.6	+0.0
Road type	68.6	76.7	69.6	76.7	+0.0
Road surface	95.7	95.4	95.0	95.0	-0.3
Transit density	26.4	44.9	45.8	49.4	+3.6
OVERALL (all topics)	48.3	58.3	54.9	75.9	

Caveat: `green_space` is leaked

green_space scores **100% even in blind mode** (no images) across all runs. The gold is always the “no parks / densely built” option text regardless of A/B/C/D position — the model answers from a **text prior**, not vision. It carries **zero vision signal** and is excluded from the fusion means above. (Worth regenerating with balanced golds.)

Appendix — ALL 14 tasks (full & fusion, every run)

task	pc-val	fus	su-val	fus	pc-bench	fus	su-bench	fus
Land use	75.2	-1.4	69.1	-0.1	75.3	-0.2	73.9	+1.4
Building height	66.3	-0.8	58.8	-0.4	66.8	-0.5	66.8	+0.5
Urban density	59.4	-2.7	70.0	+3.1	65.1	+2.4	74.6	+3.1
Junction type	74.6	+0.4	73.5	+1.5	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	96.0	-3.1	100.0	+0.0	98.0	-1.5
Amenity richness	62.4	+4.8	54.0	-0.1	58.7	+2.1	55.6	+0.0
Road type	75.2	-1.2	70.0	-2.0	76.5	-1.0	76.7	+0.0
Road surface	94.2	+0.3	98.1	+0.0	95.7	+0.7	95.0	-0.3
Transit density	50.6	+3.0	45.2	+0.9	50.1	+4.5	49.4	+3.6
Camera direction	46.5	+21.4	44.7	+20.8	42.3	+17.3	43.7	+20.9
Mismatch bin (easy)	95.2	+48.1	96.6	+47.1	97.4	+44.7	98.8	+46.1
Mismatch bin (hard)	90.4	+39.2	87.8	+37.6	90.7	+44.9	87.9	+43.7
Mismatch MCQ (easy)	95.4	+69.2	95.0	+70.8	96.2	+69.8	92.6	+67.9
Mismatch MCQ (hard)	84.3	+57.2	82.7	+57.6	86.0	+62.7	84.8	+55.1

pc=per-city, su=seen/unseen, val=own split, bench=shared benchmark. **mismatch_*** fuse strongly (+40–70)
 — they *require* both views by construction.